

Eksamen	Exam
Kopiereg voorbehou	Copyright reserved
Elektrisiteit en Elektronika EBN 122	Electricity and Electronics EBN 122
13 November 2018	13 November 2018

Toetsinligting - Test Information			
Maksimum punte: Maximum marks:	90	Volpunte: Full marks:	93
Duur van vraestel: Duration of paper:	180 min + (20 for OCR)	Oop-/Toeboek: Open/Closed book:	Toe/Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (this page included)		31	

Important	Belangrik
1. The examination regulations of the University of Pretoria apply. 2. All questions must be answered on the Optical Character Recognition (OCR) form. 3. Read the instructions on the OCR form carefully. 4. Where applicable, answers must include units. 5. Final answers must be written as real numbers and not as fractions. 6. Use at least 3 significant numbers to write irrational numbers.	1. Die eksamenregulasies van die Universiteit van Pretoria geld. 2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word. 3. Neem noukeurig kennis van die instruksies op die OKH vorm. 4. Indien van toepassing, moet antwoorde eenhede insluit. 5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie. 6. Gebruik ten minste 3 beduidende syfers om irrasionele getalle voor te stel.
Lecturers/Dosente:	D. le Roux, J.P. de Villiers, J.P. Jacobs

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 180 minutes of the exam. (0-180 min)

Do step 2 within the following 10 minutes. (180-200 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

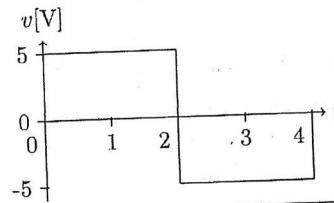
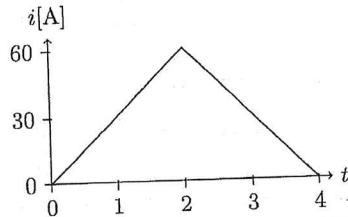
STEP 1 (0 min to 180 min): Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.

STEP 2 (180 min to 200 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN122E1**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1 / Afdeling 1

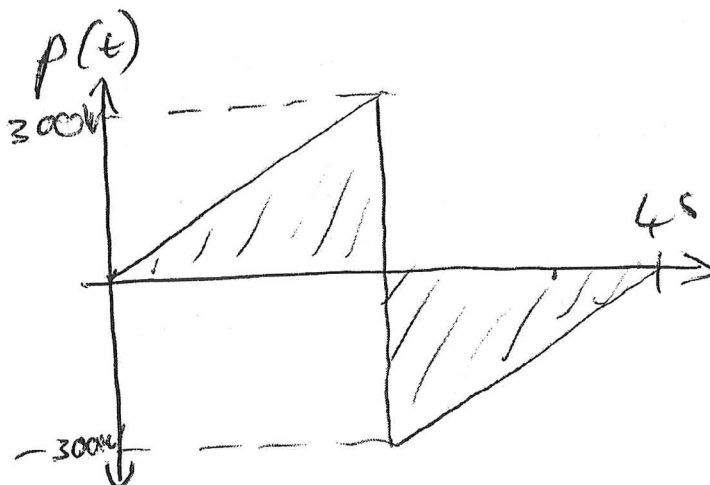
Questions 1 to 2 [2]		Vrae 1 tot 2 [2]	
The figure below shows the current through and the voltage across a circuit element.		Die figuur hieronder toon die stroom deur en die spanning oor 'n kringbaanelement.	
1	Determine the power delivered to the element at time $t = 3\text{ s}$. [1]	1	Bepaal die drywing gelewer aan die element by tyd $t = 3\text{ s}$. [1]
2	Find the total energy absorbed by the element for the period of $0 < t < 4\text{ s}$. [2]	2	Bepaal die totale energie geabsorbeer deur die element vir die tydskuur $0 < t < 4\text{ s}$. [2]



$$1. \quad p = v i \quad \text{at} \quad t = 3\text{ s}$$

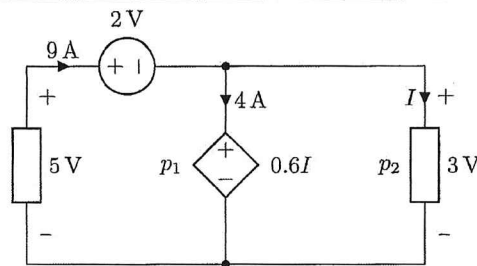
$$= -5(30) = -150\text{ W}$$

$$2. \quad w = \int_0^4 p(t) dt$$



$$w = 0\text{ J}$$

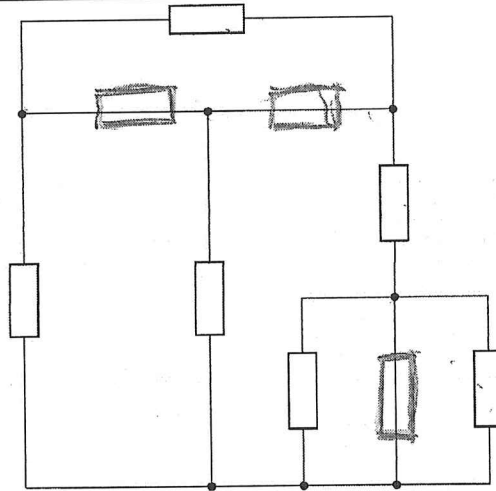
Questions 3 to 4		Vrae 3 tot 4	
Assuming $I = 5$ A, calculate the power relating to the following elements in the circuit below:		Indien aanvaar word dat $I = 5$ A, bereken die drywing geassosieer met die volgende elemente in die onderstaande stroombaan:	
3	p_1 [1]	3	p_1 [1]
4	p_2 [1]	4	p_2 [1]



$$\begin{aligned}
 3. \quad p_1 &= +V\bar{I} \\
 &= (0.6(5))4 = 12 \text{ W}
 \end{aligned}$$

$$4 \quad p_2 = +V\bar{I} = 5(3) = 15 \text{ W.}$$

5-7 Questions 7 to 9 [3]		5-7 Vrae 7 tot 9 [3]	
Determine the number of nodes, branches, and independent loops in the circuit diagram below.		Bepaal die hoeveelheid nodes, takke, en onafhanklike lusse in die kringbaan hieronder.	
5	Nodes n [1]	5	Nodusse n [1]
6	Branches b [1]	6	Takke b [1]
7	Independent loops l [1]	7	Onafhanklike lusse l [1]



5. $n = 5$

6. $b = 9$

7. $l = 5$

$$\begin{aligned}
 b &= l + n - 1 \\
 &= 4 + 5 - 1 \\
 &= 8 \quad \checkmark
 \end{aligned}$$

→

8-9 Questions 10 to 11 [2]		8-9 Vrae 10 tot 11 [2]	
Consider the circuit below. Determine:		Beskou die kringbaan hieronder. Bepaal	
8 10	I [1]	8 10	I [1]
9 11	v_o [1]	9 11	v_o [1]

$v_o = -12I$

8) KVL: $-24 + 8I + 2v_o - 8 - v_o = 0$

$$-24 + 8I + v_o - 8 = 0$$

$$8I - 12I = 32$$

$$-4I = 32$$

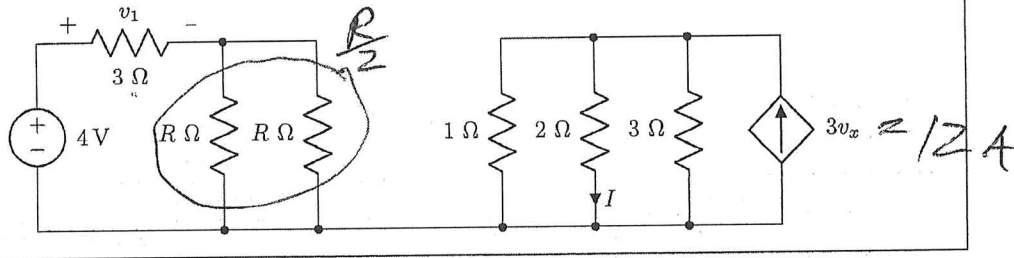
$$I = -8 \text{ A}$$



9) $v_o = -12(-8) = 96 \text{ V}$



10 to 11		10-11	
Questions 12 to 13 [3]		Vrae 12 tot 13 [3]	
Consider the circuit below.		Beskou die kringbaan hieronder.	
10 12	If $v_1 = 3$ V, determine R . [2]	10 12	Indien $v_1 = 3$ V, bepaal R . [2]
11 13	If $v_x = 4$ V, determine I . [1]	11 13	Indien $v_x = 4$ V, bepaal I . [1]



$$10. R \parallel R = \frac{1}{\frac{1}{R} + \frac{1}{R}} = \frac{1}{\frac{2}{R}} = \frac{R}{2}$$

$$v_1 = \frac{3}{3 + \frac{R}{2}} 4 = 3$$

$$3(4) = 3(3) + \frac{3R}{2}$$

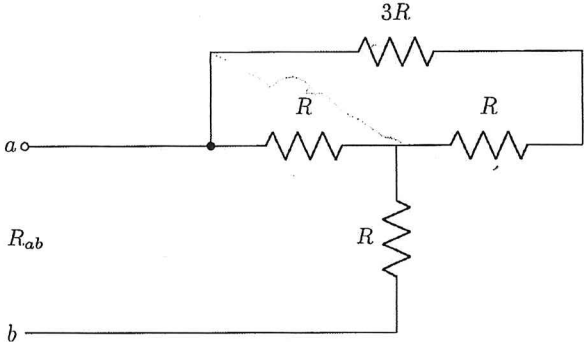
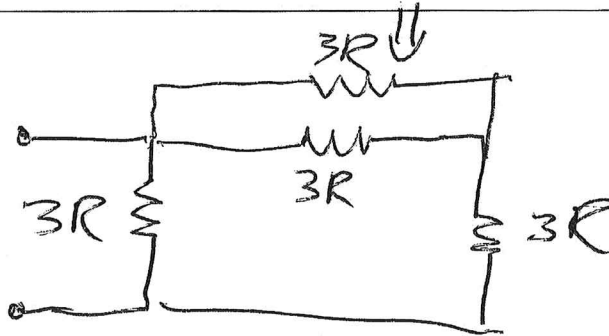
$$\frac{3R}{2} = 3 \quad \therefore 3R = 6$$

$$R = 2 \Omega \rightarrow$$

$$11. \frac{1}{3} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{3}{3} + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4} \Omega$$

$$I = \frac{\frac{3}{4}}{2 + \frac{3}{4}} 12 = 3.2727 A \rightarrow$$

Question 12 [2]		Vraag 12 [2]	
12	Determine R if $R_{ab} = 90 \Omega$.	12	Bepaal R indien $R_{ab} = 90 \Omega$.

$$R_{ab} = 3R // \left(\frac{3}{2}R + 3R \right)$$

$$\frac{1}{R_{ab}} = \frac{1}{3R} + \frac{1}{\left(\frac{3R}{2} + \frac{6R}{2} \right)} = \frac{1}{3R} + \frac{1}{\frac{9R}{2}}$$

$$= \frac{1}{3R} + \frac{2}{9R} = \frac{3}{9R} + \frac{2}{9R} = \frac{5}{9R}$$

$$R_{ab} = \frac{9R}{5} = 90 \quad \therefore 9R = 90$$

$$R = 10 \Omega$$

13-17 Questions 15 to 19 [5]		13-17 Vrae 15 tot 19 [5]	
Consider the circuit below containing a transistor with $\beta = 150$ and $V_{BE} = 0.7$ V. Use mesh analysis to set up the following two equations of the form: $ai_1 + bi_2 = 2 \times 10^{-5}$ $ci_1 + di_2 = -0.7 \times 10^{-5}$ Calculate the unknown coefficients.		Beskou die kringbaan hieronder wat 'n transistor bevat met $\beta = 150$ en $V_{BE} = 0.7$ V. Gebruik lusanalise om die volgende twee vergelykings op te stel met die vorm: $ai_1 + bi_2 = 2 \times 10^{-5}$ $ci_1 + di_2 = -0.7 \times 10^{-5}$ Bepaal die onbekende koëffisiënte.	
15	a [1]	16	b [1]
17	c [1]	18	d [1]
19	Determine the value of V_{CE} . [1] $i_2 = 9.5 \mu A$		Bepaal die waarde van V_{CE} . [1]

The circuit diagram shows a 2V DC voltage source in series with a 100 kΩ resistor. This combination is connected to a node that branches into a 200 kΩ resistor connected to ground and the base of an NPN transistor. The emitter of the transistor is connected to ground. The collector of the transistor is connected to a 1 kΩ resistor, which is in series with a 16V DC voltage source. Three mesh currents are indicated: i_1 flows clockwise through the 2V source and 100 kΩ resistor; i_2 flows clockwise through the 200 kΩ resistor and the base-emitter junction of the transistor; i_3 flows clockwise through the collector-emitter junction of the transistor, the 1 kΩ resistor, and the 16V source. The collector-emitter voltage is labeled V_{CE} .

M1 $-2 + 100k i_1 + 200k(i_1 - i_2) = 0$

$$3i_1 - 2i_2 = 2 \times 10^{-5}$$

M2 $200k(i_2 - i_1) + 0.7 = 0$

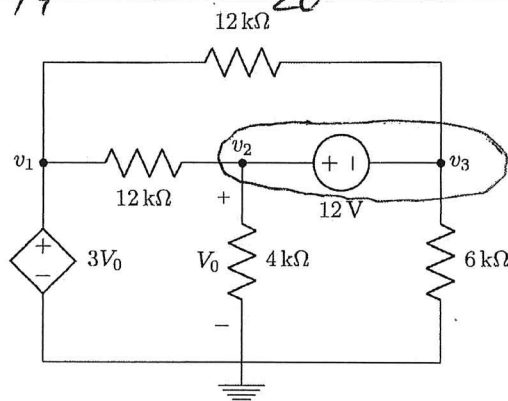
$$-2i_1 + 2i_2 = -0.7 \times 10^{-5}$$

$$a=3, b=-2, c=-2, d=2 \rightarrow$$

17) $i_3 = -150 i_2 = -1.425 \text{ mA}$

$$-V_{CE} + 1k i_3 + 16 = 0 \quad V_{CE} = -1.425 + 16 = 14.575 \text{ V} \rightarrow$$

18-21 Questions 20 to 23 [4]		18-21 Vrae 20 tot 23 [4]	
Use nodal analysis to set up the following two equations of the form: $ev_2 + 3v_3 = f$ $gv_2 + hv_3 = 12$ Calculate the unknown coefficients.		Gebruik noderale analise om die volgende twee vergelykings op te stel met die vorm: $ev_2 + 3v_3 = f$ $gv_2 + hv_3 = 12$ Bepaal die onbekende koëffisiënte.	
20 18	e [1]	21 19	f [1]
22 20	g [1]	23 21	h [1]



$$v_2 = v_3 + 12$$

$$v_1 = 3v_0 = 3v_2$$

$$\therefore v_2 - v_3 = 12 \quad (1)$$

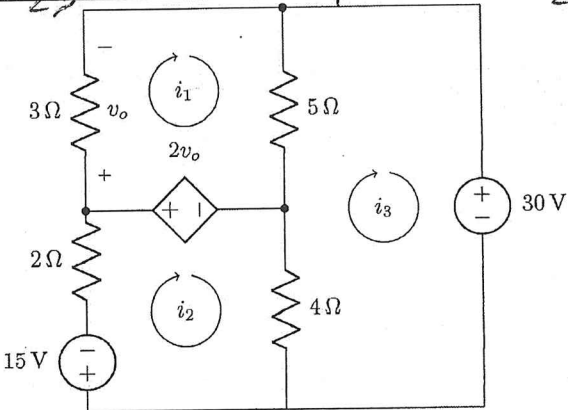
Node 2, (supernode)

$$\frac{v_2 - v_1}{12k} + \frac{v_2}{4k} + \frac{v_2 - v_1}{12k} + \frac{v_2}{6k} = 0$$

$$\times 12k \quad v_2 - 3v_2 + 3v_2 + v_3 - 3v_2 + 2v_3 = 0$$

$$-2v_2 + 3v_3 = 0$$

$$e = -2, \quad f = 0, \quad g = 1, \quad h = -1.$$

22-25 Questions 24 to 27 [4]		22-25 Vrae 24 tot 27 [4]	
Use inspection to compile a matrix equation of the form below		Gebruik inspeksie om die volgende twee matriksvergelikings op te stel met die vorm:	
$\begin{bmatrix} 2 & k & -5 \\ l & 6 & -4 \\ -5 & p & q \end{bmatrix} \mathbf{I} = \begin{bmatrix} 0 \\ -15 \\ -30 \end{bmatrix}$		$\begin{bmatrix} 2 & k & -5 \\ l & 6 & -4 \\ -5 & p & q \end{bmatrix} \mathbf{I} = \begin{bmatrix} 0 \\ -15 \\ -30 \end{bmatrix}$	
Then give the values of the following matrix entries.		Verskaf dan die waardes van die volgende matriks-elemente.	
24 k [1]	25 l [1]	26 p [1]	27 q [1]
			

$$v_o = 3i_1$$

M1 $8i_1 - 2v_o - 5i_3 = 0$

$$8i_1 - 2(3i_1) - 5i_3 = 0$$

$$2i_1 - 5i_3 = 0 \rightarrow$$

M2 $15 + 6i_2 + 2v_o - 4i_3 = 0$

$$6i_1 + 6i_2 - 4i_3 = -15$$

M3 $-5i_1 - 4i_2 + 9i_3 + 30 = 0$

$$-5i_1 - 4i_2 + 9i_3 = -30$$

$$k=0, l=6, p=-4, q=9.$$

26-27 Questions 28 to 29 [2]	26-27 Vrae 28 tot 29 [2]
Consider a transistor amplifier equivalent circuit below. $V_{BE} = 0.7$ V is the base to emitter voltage, v_{in} is the input voltage and v_{out} is the output voltage. Assume active region operation.	Beskou 'n transistorversterker ekwivalente stroombaan hieronder. $V_{BE} = 0.7$ V is die basis na emitter spanning, v_{in} is die insetspanning en v_{out} is die uitsetspanning. Aanvaar aktiewe gebied werking.
28 26 Assume that $\beta = 50$. Determine I_o if $v_{in} = 5$ V. [1]	28 26 Aanvaar dat $\beta = 50$. Bepaal I_o if $v_{in} = 5$ V? [1]
29 27 Now let $\beta = 100$. What is the output voltage v_{out} , if $v_{in} = 5$ V? [2]	29 27 Nou laat $\beta = 100$. Wat is die uitsetspanning v_{out} , indien $v_{in} = 5$ V? [2]

26) Node 1 $\frac{5 - V_i}{2k} = \frac{V_i}{4k} + \frac{V_i - 0.7}{4k}$

$$10 - 2V_i = V_i + V_i - 0.7 \therefore V_i = 2.675V$$

$$I_o = \frac{2.675 - 0.7}{4000} = 0.49375mA \rightarrow$$

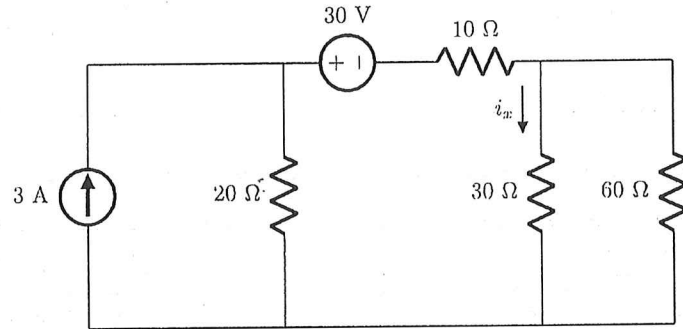
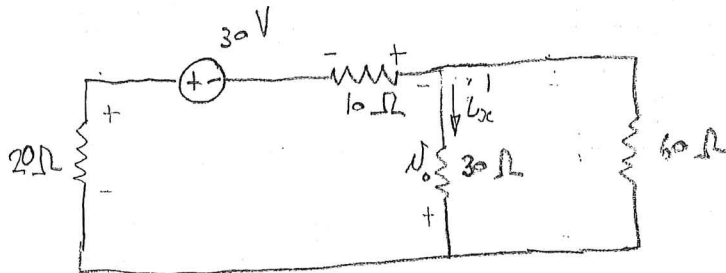
27) $\beta I_o = 49.375mA$

$$V_{out} = -(10000)(49.375mA)$$

$$= -493.75V \rightarrow$$

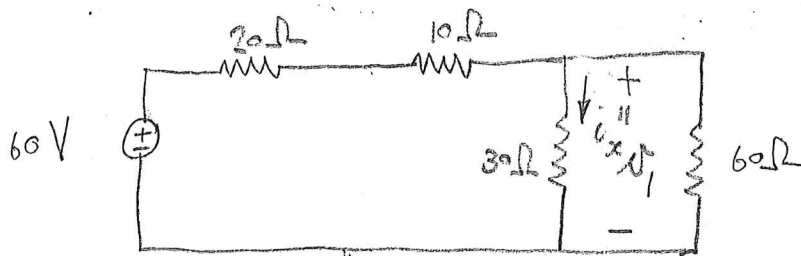
Section 2 / Afdeling 2

Questions 28 to 29 [4]		Vrae 28 tot 29 [4]	
Find the contribution to the current i_x due to:		Bepaal die bydrae tot die stroom i_x as gevolg van:	
28	the 30 V source [2]	28	die 30 V bron [2]
29	the 3 A source [2]	29	die 3 A bron [2]

28.

$$V_o = 30 \frac{30 \parallel 60}{20 + 10 + 30 \parallel 60} = 12 \text{ V}$$

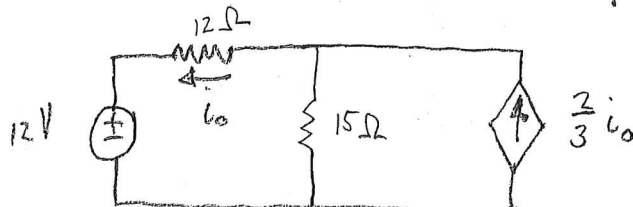
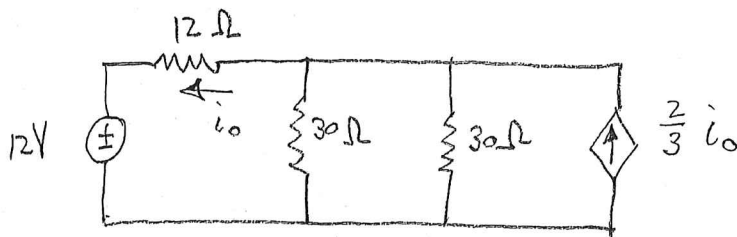
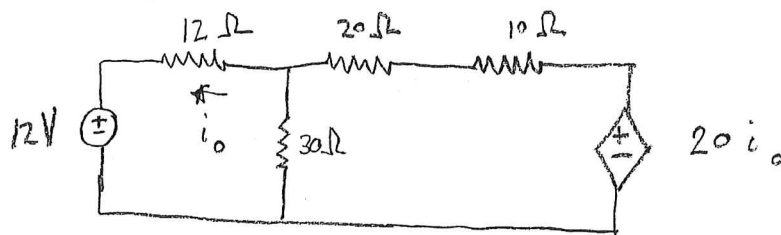
$$i_x = -\frac{V_o}{30} = -\frac{12}{30} = -0.4 \text{ A.}$$

29.

$$V_1 = 60 \frac{30 \parallel 60}{20 + 10 + 30 \parallel 60} = 24 \text{ V}$$

$$i_x = +\frac{V_1}{30} = +0.8 \text{ A}$$

Questions 30 to 32 [5]		Vrae 30 tot 32 [5]	
Simplify the top circuit by means of source transformation so that it looks like the bottom circuit. Determine:		Vereenvoudig die boonste stroombaan met behulp van brontransformasie sodat dit lyk soos die onderste stroombaan. Bepaal:	
30	R_x [2]	30	R_x [2]
31	A [1]	31	A [1]
32	B [2]	32	B [2]

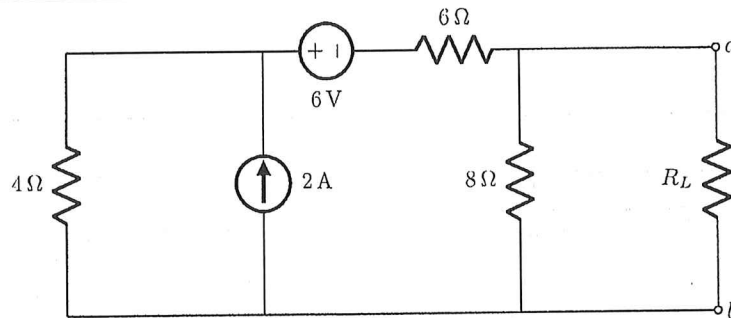


$$A = 12$$

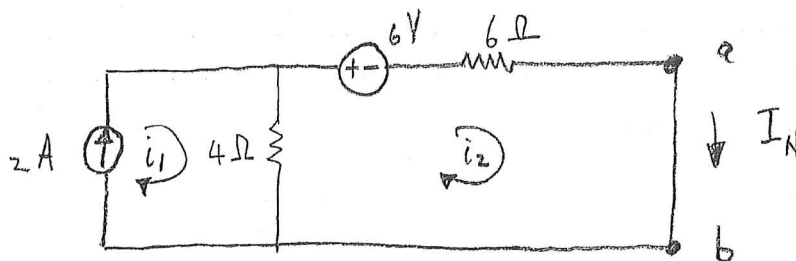
$$B = \frac{2}{3}$$

$$R_x = 15 \Omega$$

Questions 33 to 34 [4]		Vrae 33 tot 34 [4]	
Determine the Norton equivalent current I_N with reference to terminals $a - b$, and the value of the load resistor R_L so that maximum power is transferred to R_L .		Bepaal die Norton ekwivalente stroom I_N met verwysing tot terminale $a - b$, en die waarde van die lasweerstand R_L sodat maksimum drywing oorgedra word aan R_L .	
33	I_N [2]	33	I_N [2]
34	R_L [2]	34	R_L [2]



33:



Mesh 1:

$$i_1 = 2A$$

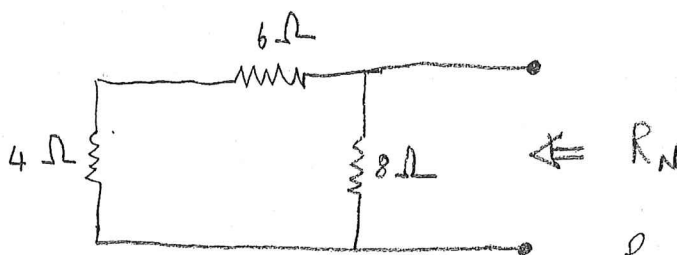
$$\text{Mesh 2: } 4(i_2 - i_1) + 6 + 6i_2 = 0$$

$$4i_2 - 8 + 6 + 6i_2 = 0$$

$$10i_2 = 2$$

$$\therefore i_2 = I_N = 0.2A$$

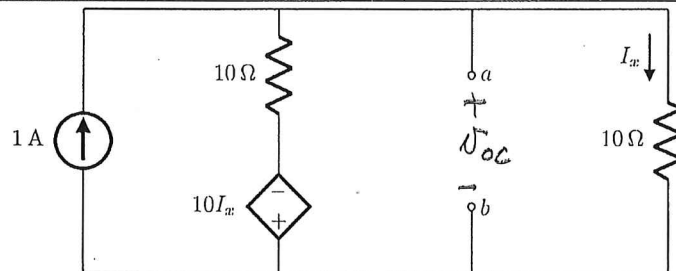
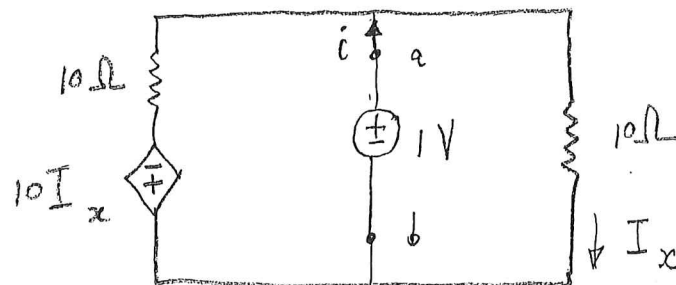
34:



$$R_L = R_N = 8 \parallel (4 + 6)$$

$$= 4.4444\Omega$$

Questions 35 to 36 [4]		Vrae 35 tot 36 [4]	
Find the Thevenin equivalent voltage V_{Th} and the Thevenin equivalent resistance R_{Th} with respect to terminals $a - b$.		Bepaal die Thevenin ekwivalente spanning V_{Th} en die Thevenin ekwivalente weerstand R_{Th} met betrekking tot terminale $a - b$.	
35	V_{Th} [2]	35	V_{Th} [2]
36	R_{Th} [2]	36	R_{Th} [2]

36 R_{Th} :

$$i = I_x + \frac{1 - (-10I_x)}{10}$$

But $I_x = \frac{1}{10}$

$$\therefore i = \frac{1}{10} + \frac{1 - (-10 \frac{1}{10})}{10} = 0.3 \text{ A}$$

$$R_{Th} = \frac{1}{i} = 3.333 \Omega$$

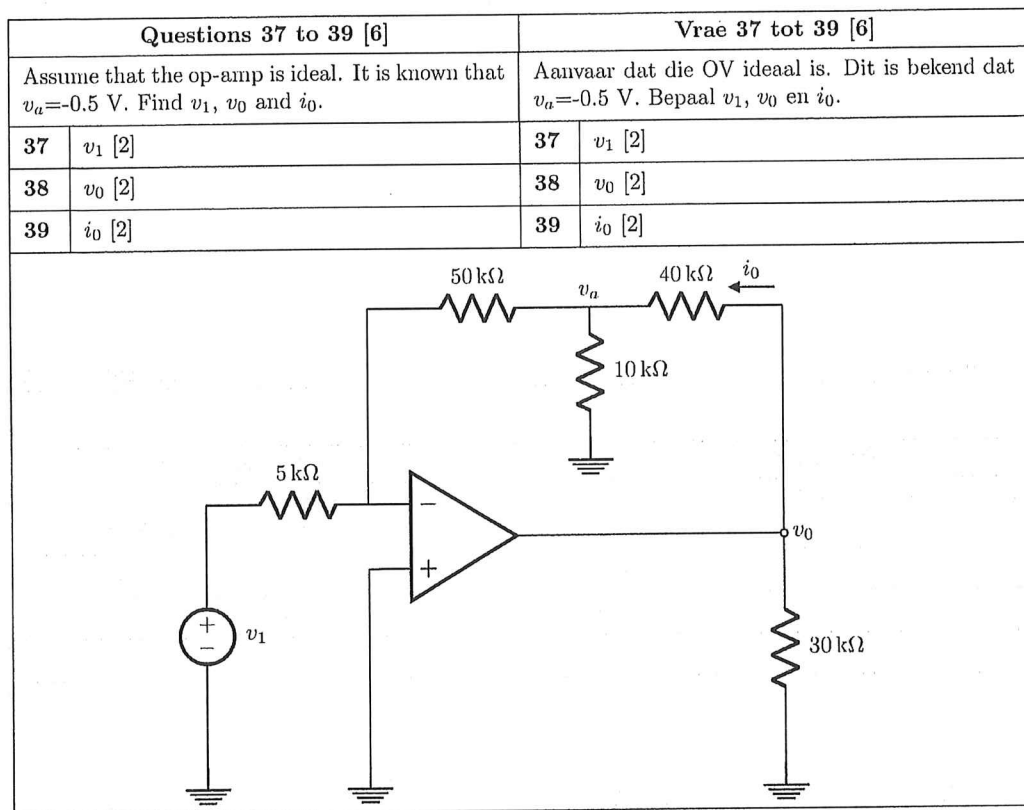
35 V_{Th} :

KCL at top : $1 = \frac{V_{oc} - (-10I_x)}{10} + I_x$

But $I_x = \frac{V_{oc}}{10}$

$$\therefore 1 = \frac{V_{oc} - (-10 \frac{V_{oc}}{10})}{10} + \frac{V_{oc}}{10}$$

$$= 0.3 V_{oc} \quad \therefore V_{Th} = V_{oc} = 3.333 \text{ V}$$



v_1 :
$$\frac{v_1 - 0}{5000} = \frac{0 - v_a}{50000}$$

$$v_1 = -0.1 v_a = -0.1 (-0.5) = 50 \text{ mV} \rightarrow$$

v_0 :
$$\frac{0 - v_a}{50000} = \frac{v_a}{10000} + \frac{v_a - v_0}{40000}$$

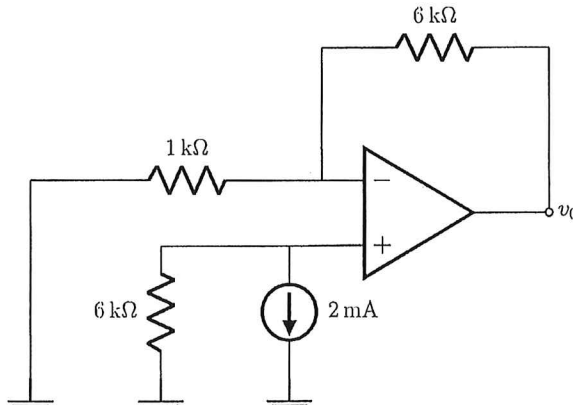
$$-v_a = 5v_a + 1.25v_a - 1.25v_0$$

$$1.25v_0 = 7.25v_a = 7.25(-0.5)$$

$$\therefore v_0 = -2.9 \text{ V} \rightarrow$$

i_0 :
$$i_0 + \frac{0 - v_a}{50000} = \frac{v_a}{10000}$$

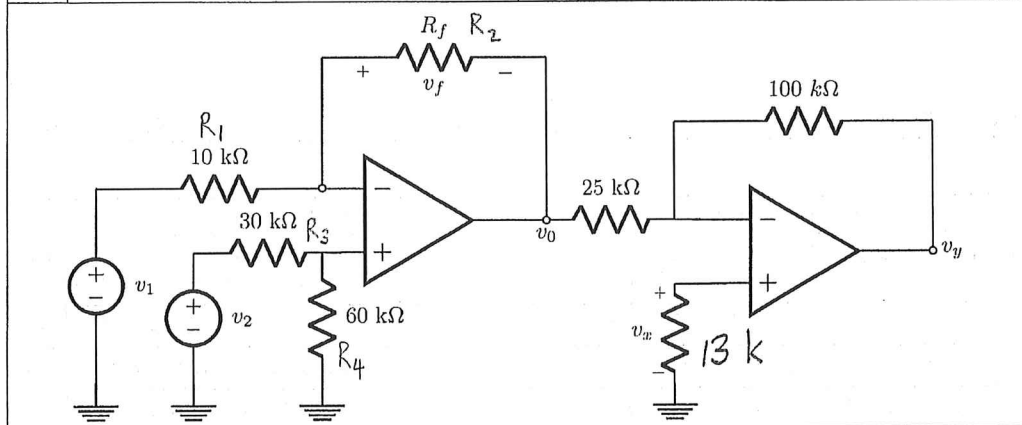
$$\therefore i_0 = \frac{v_a}{10000} + \frac{v_a}{50000} = -60 \mu\text{A} \rightarrow$$

Question 40 [2]		Vraag 40 [2]	
Assume that the op-amp is ideal. Find v_0 .		Aanvaar dat die OV ideaal is. Bepaal v_0 .	
40	v_0 [2]	40	v_0 [2]
			

$$\begin{aligned}
 v_o &= \left(1 + \frac{R_f}{R_s}\right) v_i \\
 &= \left(1 + \frac{6000}{1000}\right) (-6000 \cdot 2 \times 10^{-3}) \\
 &= -84 \text{ V}
 \end{aligned}$$

→

Questions 41 to 43 [6]		Vrae 41 tot 43 [6]	
Assume that the op-amps are ideal, and that $v_2=12$ V. It is known that $v_0=0$ when $v_1=v_2$.		Aanvaar dat die OV's ideaal is en dat $v_2=12$ V. Dit is bekend dat $v_0=0$ as $v_1=v_2$.	
41	Find R_f . [2]	41	Bepaal R_f . [2]
42	Assume that $v_0=20$ V. Find v_f . [2]	42	Aanvaar dat $v_0=20$ V. Bepaal v_f . [2]
43	Assume that $v_0=20$ V. Find v_y . [2]	43	Aanvaar dat $v_0=20$ V. Bepaal v_y . [2]



R_f : Difference amplifier

$$\therefore \frac{10\,000}{R_f} = \frac{30\,000}{60\,000} \Rightarrow R_f = 20\,k\Omega$$

v_f :

$$v_a - v_f - v_o = 0$$

$$12 \frac{60}{30+60} - v_f - 20 = 0 \quad \therefore v_f = -12\,V$$

v_y :

$$v_y = - \frac{100\,000}{25\,000} v_o$$

$$= -80\,V$$

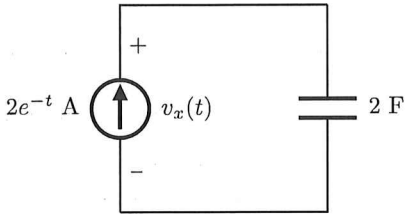
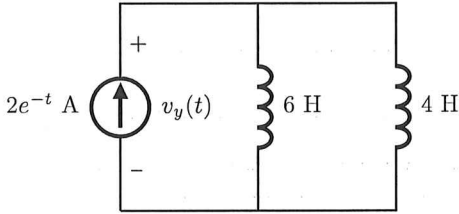
Total Section 2 / Totaal Afdeling 2: [31]

Check: $\frac{v_1 - 8}{10\,000} = \frac{8 - 20}{20\,000} \quad v_1 - 8 = 4 - 10 \therefore v_1 = 2\,V$

$$v_o = \frac{R_2}{R_1} (v_2 - v_1) = 2 (12 - 2) = 20\,V$$

Section 3 / Afdeling 3

Questions 44 to 45 [4]		Vrae 44 tot 45 [4]	
Calculate the voltages $v_x(t = 1\text{ s})$ and $v_y(t = 1\text{ s})$. Assume that $v_x(0) = 0\text{ V}$ and $v_y(0) = -1\text{ V}$. -1 V -6.8 V		Bereken die spannings $v_x(t = 1\text{ s})$ en $v_y(t = 1\text{ s})$. Aanvaar $v_x(0) = 0\text{ V}$ en $v_y(0) = -1\text{ V}$. -1 V -6.8 V	
44	$v_x(t = 1\text{ s})$ [2]	44	$v_x(t = 1\text{ s})$ [2]
45	$v_y(t = 1\text{ s})$ [2]	45	$v_y(t = 1\text{ s})$ [2]

$$i_c(t) = C \frac{dv_x(t)}{dt}$$

$$\therefore v_x(1) = \frac{1}{C} \int_0^1 i_c(t) dt + v_x(0)$$

$$= \frac{1}{2} \int_0^1 2e^{-t} dt + 0$$

$$= -e^{-t} \Big|_0^1$$

$$= 1 - e^{-1}$$

$$= \underline{0.632\text{ V}}$$

$$2e^{-t} \text{ A} \quad \uparrow \quad v_y \quad \text{6 H} \quad \text{4 H} \quad 6\text{ H} = 2.4\text{ H}$$

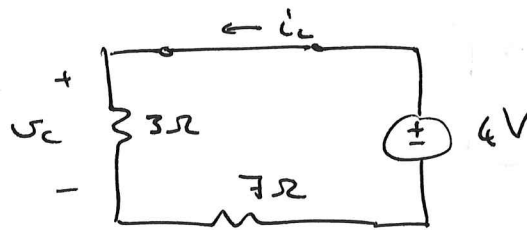
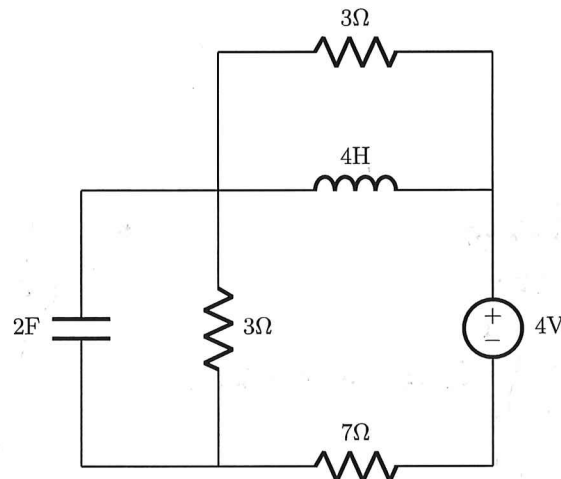
$$v_y = L \frac{di}{dt}$$

$$= 2.4 \frac{d}{dt} 2e^{-t}$$

$$= -4.8 e^{-t}$$

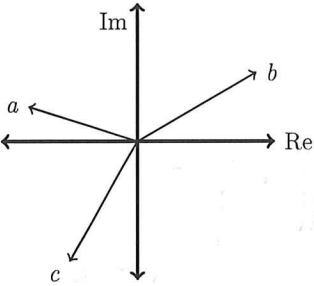
$$v_y(1) = \underline{-1.77\text{ V}}$$

Questions 46 to 47 [3]		Vrae 46 tot 47 [3]	
Under direct current conditions, calculate the energy stored in the capacitor (w_C) and the energy stored in the inductor (w_L).		Onder gelykstroomtoestande, bepaal die energie gestoor in die kapasitor (w_C) en die energie gestoor in die induktor (w_L).	
46	w_C [2]	46	w_C [2]
47	w_L [1]	47	w_L [1]



$$* \quad i_L = \frac{4}{3+7} = 0.4 \text{ A} \quad \rightarrow \quad w_L = \frac{1}{2} (4) (0.4)^2 = \underline{0.32 \text{ J}}$$

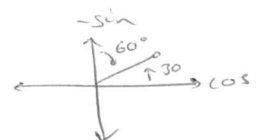
$$* \quad U_C = 3 \times 0.4 = 1.2 \text{ V} \quad \rightarrow \quad w_C = \frac{1}{2} (2) (1.2)^2 = \underline{1.44 \text{ J}}$$

Question 48 [2]	Vraag 48 [2]
Given the diagram below, which statement is correct? 1. Phasor b lags a and lags c. 2. Phasor c lags b and leads a. ✓ 3. Phasor c leads b and leads a. 4. Phasor b leads a and lags c. (Write down the number of the correct statement as your answer.)	Gegewe onderstaande diagram, watter stelling is korrek? 1. Fasor b loop na a en na c. 2. Fasor c loop na b en voor a. ✓ 3. Fasor c loop voor b en voor a. 4. Fasor b loop voor a en na c. (Skryf die nommer neer van die korrekte stelling as jou antwoord.)
48 Statement #:	48 Stelling #:
	

Question 49 [2]	Vraag 49 [2]
The following sinusoids are given: $v_1(t) = -5 \sin(\omega t - 60^\circ) \text{ V}$ $v_2(t) = 4 \cos(\omega t - 180^\circ) \text{ V}$ $v_3(t) = 3 \sin(\omega t + 60^\circ) \text{ V}$ $\bar{V}_1$, $\bar{V}_2$ and $\bar{V}_3$ are the phasor representations of $v_1(t)$, $v_2(t)$ and $v_3(t)$ respectively. Determine the phasor value of $\bar{V}_T = \bar{V}_1 + \bar{V}_2 + \bar{V}_3$. Give the answer in polar form.	Die volgende sinusiede word gegee: $v_1(t) = -5 \sin(\omega t - 60^\circ) \text{ V}$ $v_2(t) = 4 \cos(\omega t - 180^\circ) \text{ V}$ $v_3(t) = 3 \sin(\omega t + 60^\circ) \text{ V}$ $\bar{V}_1$, $\bar{V}_2$ and $\bar{V}_3$ is die fasorvoorstellings van $v_1(t)$, $v_2(t)$ and $v_3(t)$ respektiewelik. Bepaal die fasor waarde van $\bar{V}_T = \bar{V}_1 + \bar{V}_2 + \bar{V}_3$. Gee die antwoord in polêre vorm.
49 $\bar{V}_T$	49 $\bar{V}_T$

$$v_1 = -5 \sin(\omega t - 60^\circ)$$

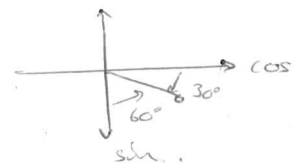
$$= 5 \cos(\omega t + 30^\circ) \rightarrow V_1 = 5 \angle 30^\circ \text{ V}$$



$$v_2 = 4 \cos(\omega t - 180^\circ) \rightarrow +V_2 = 4 \angle -180^\circ \text{ V}$$

$$v_3 = 3 \sin(\omega t + 60^\circ)$$

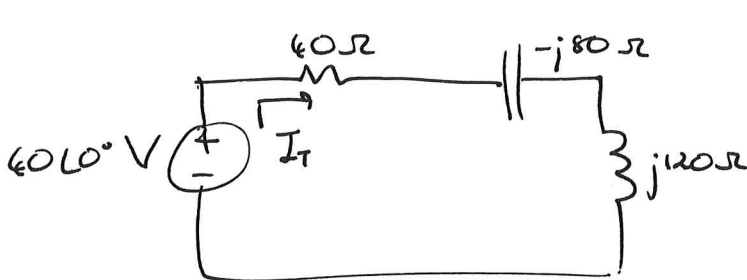
$$= 3 \cos(\omega t - 30^\circ) \rightarrow +V_3 = 3 \angle -30^\circ \text{ V}$$



$$V_T = 4.068 \angle 35.2^\circ \text{ V}$$

$$3.09 \angle 18.9^\circ \text{ V}$$

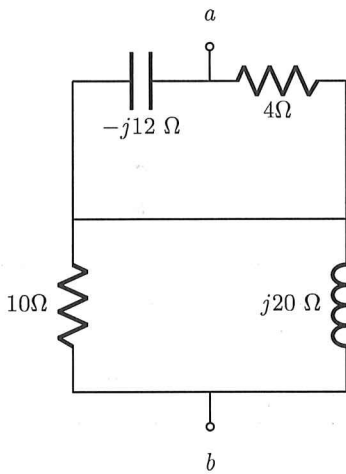
Question 50 [2]	Vraag 50 [2]
A series circuit consists of a $40 \cos(5 \times 10^3 t)$ V voltage source, a 40Ω resistor, a $2.5 \mu\text{F}$ capacitor, and 24 mH inductor. The current $i_T(t)$ exits the positive terminal of the voltage source. $\bar{I}_T$ is the phasor representation of $i_T(t)$. Calculate the current $\bar{I}_T$. Give the answer in rectangular form.	'n Serie-kringbaan bestaan uit 'n $40 \cos(5 \times 10^3 t)$ V spanningsbron, 'n 40Ω resistor, 'n $2.5 \mu\text{F}$ kapasitor, en 'n 24 mH induktor. Die stroom $i_T(t)$ vloei uit die positiewe terminaal van die stroombron. $\bar{I}_T$ is die fasorvoorstelling van $i_T(t)$. Bepaal die stroom $\bar{I}_T$. Gee die antwoord in reghoekige vorm.
50 $\bar{I}_T$	50 $\bar{I}_T$



$$Z_C = \frac{-j}{5000 \times 2.5 \mu} = -j80 \Omega$$

$$Z_L = j5000 \times 24 \text{ m} = j120 \Omega$$

$$\begin{aligned} \bar{I}_T &= \frac{40}{40 + j40} \\ &= \underline{0.5 - j0.5 \text{ A}} \end{aligned}$$

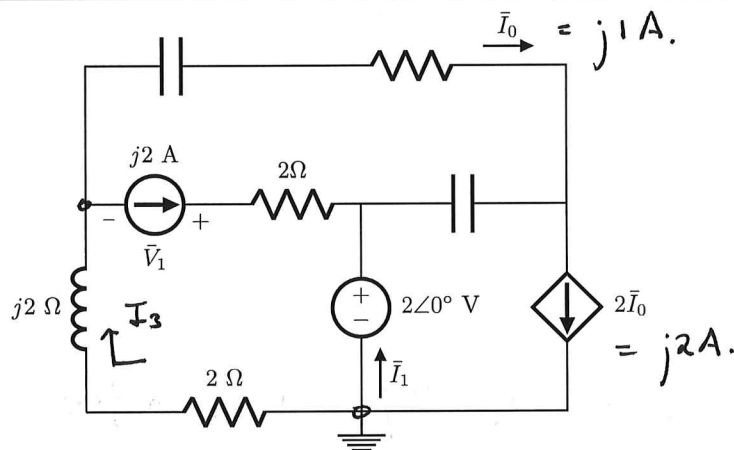
Question 5f [2]		Vraag 5f [2]	
Determine Z_{ab} . Give the answer in rectangular form .		Bepaal Z_{ab} . Gee die antwoord in reghoekige vorm .	
5f	Z_{ab}	5f	Z_{ab}
			

$$Z_{ab} = (10 \parallel j20) + (4 \parallel -j12)$$

$$= \frac{10 \times j20}{10 + j20} + \frac{-j12 \times 4}{4 - j12}$$

$$= \underline{11.6 + j2.8 \, \Omega}$$

Questions 52 to 53 [4]		Vrae 52 tot 53 [4]	
Given: $\bar{I}_0 = 1\angle 90^\circ$ A. Determine $\bar{V}_1$ and $\bar{I}_1$. Give the answers in rectangular form.		Gegee: $\bar{I}_0 = 1\angle 90^\circ$ A. Bepaal $\bar{V}_1$ en $\bar{I}_1$. Gee die antwoorde in reghoekige vorm.	
52	$\bar{V}_1$	52	$\bar{V}_1$
53	$\bar{I}_1$	53	$\bar{I}_1$



* KCL: $I_3 = j2 + j1 = j3$ A.

KCL: $j2 = I_1 + I_3$

$\therefore I_1 = \underline{-j1}$ A.

* KVL: $-V_1 + 2(j2) + 2 + (2 + j2) I_3 = 0$

$V_1 = j4 + 2 + j6 - 6$

$= \underline{-4 + j10}$ V

Questions 54 to 57 [4]		Vrae 54 tot 57 [4]	
Use nodal analysis to set up the following two equations of the form: $a\bar{V}_1 + b\bar{V}_2 = j19$ $c\bar{V}_1 + d\bar{V}_2 = -j84$ Calculate the unknown coefficients and write in rectangular format.		Gebruik nodelanalyse om die volgende twee vergelykings op te stel met die vorm: $a\bar{V}_1 + b\bar{V}_2 = j19$ $c\bar{V}_1 + d\bar{V}_2 = -j84$ Bepaal die onbekende koëffisiënte en skryf in reghoekige formaat.	
54	a [1]	55	b [1]
		56	c [1]
		57	d [1]

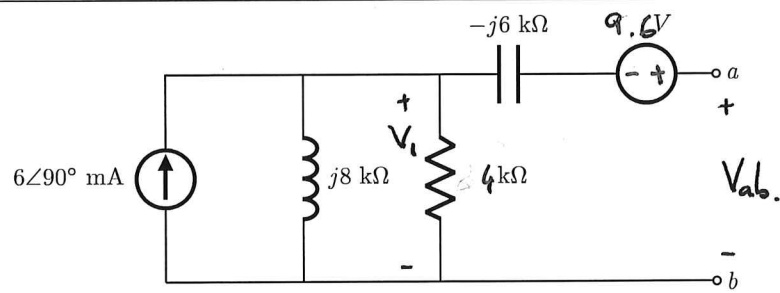
$$\textcircled{1} \quad \frac{V_1 - 3}{8} + \frac{V_1}{-j4} + \frac{V_1 - j4 - V_2}{j2} = 0.$$

$$\begin{aligned} (\times j8) \quad jV_1 - j3 + (-2V_1) + 4V_1 - j16 - 4V_2 &= 0. \\ V_1(2+j) + V_2(-4) &= j19 \end{aligned}$$

$$\textcircled{2} \quad \frac{V_2}{2} + \frac{V_2 + j4 - V_1}{j2} = j2V_1.$$

$$\begin{aligned} (\times j2) \quad jV_2 + V_2 + j4 - V_1 - j2V_1 &= 0 \\ \cancel{V_1(-1-j2)} + V_2(1+j) &= -j4 \end{aligned}$$

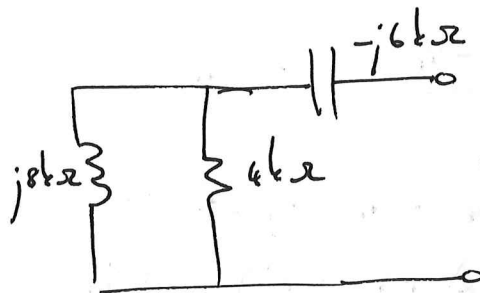
$$\begin{aligned} (\times j2) \quad jV_2 + V_2 + j4 - V_1 &= -4V_1 \\ 3V_1 + V_2(1+j) &= -j4 \end{aligned}$$

Questions 58 to 59 [4]		Vrae 58 tot 59 [4]	
Determine the Thevenin equivalent voltage and impedance for the circuit below with respect to terminals a-b. Give the answers in rectangular form.		Bepaal die Thevenin ekwivalente spanning en impedansie vir onderstaande kringbaan met verwysing tot terminale a-b. Gee die antwoorde in reghoekige vorm.	
58	$\bar{V}_{TH}$	58	$\bar{V}_{TH}$
59	Z_{TH}	59	Z_{TH}
			

 V_{TH}

$$\begin{aligned}
 * V_1 &= 6\angle 90^\circ \text{ mA} \times (j8 \parallel 4 \text{ k}) \\
 &= j6 \text{ mA} \times \frac{(j8)(4) \text{ k}}{j8 + 4} \\
 &= -9.6 + j19.2 \text{ V.}
 \end{aligned}$$

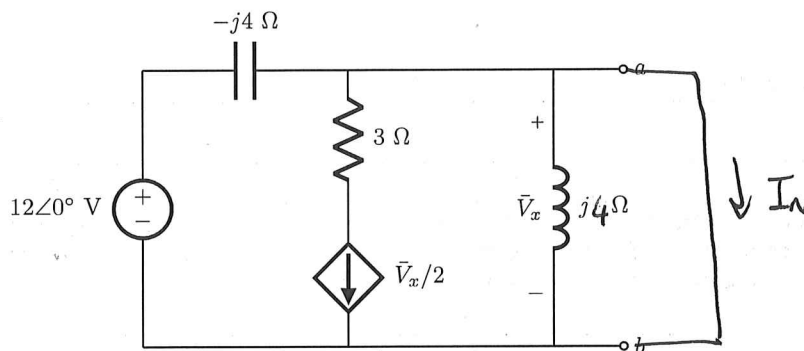
$$* V_{ab} = V_1 + 9.6 = \underline{0 + j19.2 \text{ V}}$$

 Z_{TH} 

$$\leftarrow Z_{ab} = (-j6 + j8 \parallel 4) \text{ k}.$$

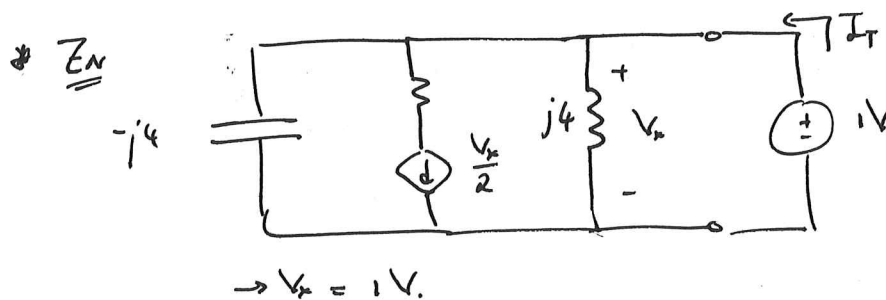
$$= \underline{3.2 - j4.4 \text{ k}\Omega}$$

Questions 60 to 61 [4]	Vrae 60 tot 61 [4]
Determine the Norton equivalent current and impedance for the circuit below with respect to terminals a-b. Give the answers in rectangular form .	Bepaal die Norton ekwivalente stroom en impedansie vir onderstaande kringbaan met verwysing tot terminale a-b. Gee die antwoorde in reghoekige vorm .
60 $\bar{I}_N$	60 $\bar{I}_N$
61 Z_N	61 Z_N



* $\underline{I_N}$ $\rightarrow V_x = 0V. \therefore$

$\therefore I_N = \frac{12}{-j4} = j3 A$



Nodal $\rightarrow I_T = \frac{1}{j4} + \frac{1}{2} + \frac{1}{-j4} \quad \therefore Z_{TH} = \frac{1}{I_T} = 2\Omega$

$= 0.5 + j0$

Total Section 3 / Totaal Afdeling 3: [31]

Grand Total / Groottotaal : [93]

